

WASP-43b

R-1786

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_210212 · created 2026-08-03T00:09:38+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459257.8336 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.145 +/- 0.026
Transit depth (Rp/Rs)^2	2.11 +/- 0.77 [%]
Orbital Inclination (inc)	80.17 +/- 2.13
Airmass coefficient 1 (a1)	0.8178 +/- 0.0084
Airmass coefficient 2 (a2)	0.144 +/- 0.042
Scatter in the residuals of the lightcurve fit is	1.03 %
Best Comparison Star	#1 - [387, 43]
Optimal Method	PSF photometry
Transit Duration (day)	0.041 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.145 ± 0.026 vs 0.1594 (deviation 0.55σ)
Duration fit vs published	0.041 d vs 0.0512 d (deviation 0.85σ)
Mid-time O–C	-0.3 min
Depth SNR	5.6
Residual scatter	1.03 %

Light curve

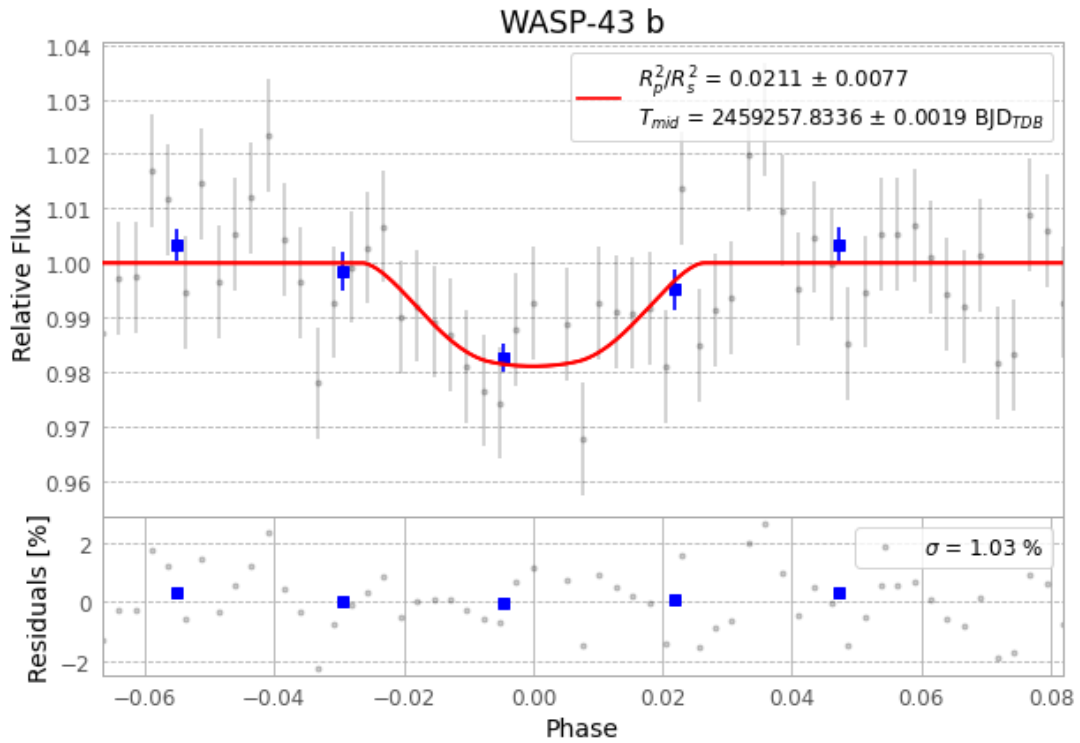


Figure 1 — FinalLightCurve_WASP-43 b_2021-02-11.png (SHA-256 db8173c933bea74c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [462, 238], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ac6323de4db18960...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

59 FITS frames (39 MB), WASP-43210212063312.FITS → WASP-43210212092712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-210212.log (95e1157b200a...), FinalLightCurve_WASP-43 b_2021-02-11.png (db8173c933be...), FinalLightCurve_WASP-43 b_2021-02-11.csv (c1b2895f1f4b...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 00:09Z.